

BUSINESS ANALYST INTERN TECHNICAL ASSESSMENT

Question Set: A

Congratulations on clearing the aptitude round. This assessment has four sections, all built on a single CSV. Section 2 is a substantial CASE STUDY read its briefing carefully. The mini-project (Section 4) is the only open-ended deliverable; everything else has specific, verifiable answers.

Suggested time: 4 hours, submit a single ZIP. Read all footnotes before starting they document real data quirks that will sink surface-level analysis.

The Dataset

A single CSV (hotel bookings.csv) of 12,000 booking records from a hotel-booking platform. Each row is one booking; customer and property attributes are denormalized into the row. 28 columns:

- **Booking id, customer id, property id** — Primary key + foreign keys
- **Customer name, customer segment, customer signup date, customer home city, customer loyalty tier** — Customer attributes (denormalized into every row)
- **Property name, property city, property star rating, property type, property total rooms** — Property attributes (denormalized into every row)
- **Booking date, check-in date, checkout date** — Key dates
- **Room type, num rooms, nights** — Room details
- **Booking channel** — Direct Website / OTA / Travel Agent / Corporate Portal
- **adr, discount amount, coupon code, total amount, payment method** — Pricing & payment (total amount = $\text{adr} * \text{nights} * \text{num rooms} - \text{discount amount}$)
- **Booking status** — Completed / Cancelled / No-Show
- **Review rating, review date** — Review info (null if no review left)

Important Footnotes read before starting

Footnote 1 Invalid stays: Some rows have checkout date on or before check-in date. Treat as data errors.

Footnote 2 Booking before account: Some rows have booking date earlier than customer signup date - a temporal-integrity issue.

Footnote 3 Zero rooms: Some rows have num rooms = 0 (carry total amount = 0). These are erroneous.

Footnote 4 Property names repeat across cities: A handful of property name values appear in multiple cities with DIFFERENT property ids. NEVER group on property name - always use property id.

Footnote 5 Reviews on Cancelled bookings: A small number of Cancelled bookings carry a review rating, which is impossible. Flag these.

Footnote 6 Review scale mismatch: review rating is on 1-5 for Individual and Group customers, but on 1-10 for corporate customers. Without normalization, any cross-segment rating average is wrong.

Footnote 7 customer loyalty tier 'None' parsing trap: The tier column has 4 valid values including the literal string 'None' (meaning 'no tier'). Default CSV readers (pandas read csv, etc.) will silently convert 'None' to a missing value, making ~46% of your tier data disappear. Use keep default Na=False (pandas) or the equivalent in your tool.

Footnote 8 Realized vs. booked revenue: Only booking status = 'Completed' counts as realized revenue. Cancelled and No-Show rows still carry a total amount (the would-have-been-charged amount); summing all rows overstates revenue by ~30%.

Section 1 Data Quality Checks (quick)

Three fast factual checks. Each answer is a specific number or short identification — enter it in the box.

A1. How many bookings have invalid stays (checkout date $\leq$ check-in date)? (integer)

A2. Compute review rating range and mean SEPARATELY for each customer segment. What does this reveal? (1-line implication)

A3. After applying the relevant data-quality footnotes, what is realized revenue for property type = Luxury? (₹)

Section 2 Case Study: The Cancellation Crisis

Briefing: The Operations head has flagged that the platform's overall booking cancellation rate is around 19% — well above the industry benchmark of 12%. The board has set a target to reduce platform-wide cancellation by 5 percentage points (to ~14%) within two quarters. They expect you to identify where the problem is concentrated, diagnose the underlying cause from the data, and recommend a targeted intervention with a quantified expected impact. A vague “we should improve our policies” will not pass. Numbers and counterfactuals will.

What we expect: You must include at least TWO labelled visualizations: one for the landscape view (Q1) and one for the diagnosis (Q3 or Q4). Every recommendation must come with explicit math — if a number isn't shown, it doesn't count.

A1 Cancellation Landscape (visualization required).

Build a clear visual of cancellation rate broken down by AT LEAST TWO dimensions simultaneously (e.g., property city $\times$ calendar month, OR booking channel $\times$ lead-time (days till check-in since booking) bucket as a grouped bar chart). State which dimension(s) drive the most variance in cancellation rate in other words, which slice of the data most strongly differentiates high-cancel from low-cancel bookings.

A2 Rate vs. Volume.

Rank slices two ways: (i) the TOP 3 slices by cancellation RATE (worst-performing), and (ii) the TOP 3 slices by absolute cancellation COUNT. Are the high-rate slices the same as the high-volume slices? In 2-3 lines, explain why this distinction matters for the board's 5pp (percentage point) target.

A3 Root Cause: Hypothesis Testing.

For your single worst-rate slice, test candidate explanations against the data, side-by-side: (i) it's a lead-time effect (compare lead-time distribution of the slice vs the rest of the platform); (ii) it's a channel-mix effect (compare channel mix of the slice vs the rest); (iii) it's a genuine city-and-season combination (compare the slice's rate against the same city in OTHER months AND a different city in the SAME months). For each hypothesis, show the comparison numbers. State which hypothesis the data best supports and how you ruled the others out.

A4 Targeted Recommendation.

Propose ONE specific, data-grounded intervention. The proposal must specify (a) WHO is targeted (city, channel, segment, lead-time, etc. be precise), (b) WHAT the policy is (e.g., non-refundable deposit, partial-refund tier, weather-indexed insurance), and (c) the QUANTIFIED expected impact in pp of platform-wide cancel rate, stated as a range. Name one risk / collateral effect.

Section 3 SQL Challenge

Tests SQL fluency AND schema understanding. The dataset is denormalized; your first task is to design a normalized schema. The queries are written against the schema YOU design.

Schema Design Task (worth as much as the queries):

Design a normalized schema by writing CREATE TABLE statements for at least three tables (customers, properties, bookings, optionally reviews). For every column specify:

- SQL data type (INTEGER, VARCHAR(n), DECIMAL(p,s), DATE, etc.) think precision for money and length for free text;
- primary key, foreign key, or neither plus NOT NULL where appropriate;
- ONE non-trivial CHECK or UNIQUE constraint addressing one of the footnotes (e.g., CHECK(checkout date > check-in date ; UNIQUE(property name, property city)

SQL Queries (against your normalized schema, or against the flat CSV loaded as one table state which):

A-Q1.

For each property city, return the property id, property name, and revenue of the property with the HIGHEST realized revenue. Use RANK () OVER (PARTITION BY property city ORDER BY revenue DESC). Group correctly given Footnote.

A-Q2.

Using LAG () over each customer's Completed bookings ordered by check-in date, compute the gap in days between consecutive bookings per customer. Return the COUNT of customers whose AVERAGE gap is under 30 days.

Section 4 Mini-Project (build something + integrate a free API)

The only open-ended part. Build a small WORKING tool or notebook that enriches the dataset using a live, free, no-key public API and produces one concrete, quantified insight. You MAY (and should) use AI assistants — we expect it. But the deliverable must be a working artifact YOU wired together, debugged, and can explain. We are assessing how well you USE AI to BUILD, not whether AI can answer a prompt.

Set A Weather-Cancellation Analyzer

API: Open-Meteo Historical Weather API free, no key. Endpoint: https://archive-api.open-meteo.com/v1/era5?latitude=&longitude=&start_date=&end_date=&daily=precipitation_sum%2Ctemperature_2m_max

What to build: For a sensible sample of bookings (all Goa and Manali check-in dates in 2024), fetch the actual weather on the check-in date and test whether heavy-rain or extreme-temperature days are associated with higher cancellation or no-show rates. Map each city to coordinates yourself and batch your calls sensibly (do not hit the API once per row group by city + date).

Every mini-project must include ALL of the following:

1. A real call to the assigned free API, with error handling timeouts or missing data should NOT crash the run. Show this in the code.
2. A correct merge of the API data back onto the bookings (watch join keys, dates, city country/coords mappings).
3. ONE non-obvious, quantified insight the dataset ALONE could not produce a number or chart, stated in 3-4 sentences.
4. A 6-line README: what it does, how to run it, one design decision, one limitation.
5. A specific note (≤ 8 lines) on where you used AI and what you had to FIX in the AI's output vague notes lose marks; specific notes win.

Deliverables & How We Grade

Submit a single ZIP containing:

1. answers.pdf (or .docx): Sections 1-3, your CREATE TABLE statements + index justification, your SQL queries pasted as text, and the case-study answers with embedded visualizations.
2. code/ folder: your analysis notebook/script + SQL files.
3. project/ folder: working mini-project code + README + the insight + AI-usage note.